Aaron Wang

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Education

University of Illinois Chicago Chicago, IL — August 2022 – Present
Ph.D., Physics

University of Washington Seattle, WA — September 2018 – June 2022
B.S., Comprehensive Physics

Awards

DOE Office of Science Graduate Student Research (SCGSR) Award – funds thesis research at a DOE national lab — 2027

URA Visiting Scholar – Universities Research Association award for research at Fermilab — 2026

Chicagoland Computational Traineeship in High Energy Particle Physics (C2 the P2) – two years of full PhD funding for work at Fermilab — 2023 – 2025

IRIS-HEP Fellowship – scientific computing software for the particle physics community — 2021

Washington Research Foundation Nanofabrication Fellowship – quantum computing research — 2019

Research Experience

Amazon Web Services, Fundamental AI Research New York, NY — May 2026 – Present
Applied Scientist Intern

- Research whether LLM agents can follow a wall-clock time budget on long-horizon tasks and use that time effectively, including time spent in tool calls and code execution that the model cannot measure from its own output
- Built time-budgeted evaluations in Factorio and Kaggle machine-learning competitions (MLE-bench) that measure deadline adherence, how much of the budget agents use, and whether they change strategy when the budget changes
- Found that agents told only the deadline stop early, using about a quarter of their time; showing them a live clock raised task output 77% with no missed deadlines
- Developed a reinforcement learning method (GRPO policy optimization) that trains agents to follow the budget and choose a different strategy for short and long budgets (e.g. in Factorio, build less and harvest sooner when the budget is short)
- Introduced trajectory grafting, a training-data augmentation that splices together agent runs from different budgets; it nearly doubled task output under long budgets on unseen maps
- Encoded wall-clock time directly in the model: repurposed part of the rotary position embedding to rotate with elapsed seconds instead of token position, so the model can attend to time the way it attends to sequence order; raised on-time task completion from 6% to 50%
- Scaled training to multi-node GPU clusters with hundreds of parallel environments, including a 27B-parameter model trained across AWS regions
- First-author paper at the Resource-Aware Agentic AI Workshop @ NeurIPS 2026

Fermi National Accelerator Laboratory Batavia, IL — June 2023 – Present
Graduate Research Fellow

- Member and author of the CMS Collaboration at the CERN Large Hadron Collider
- Design efficient transformer architectures for real-time particle jet classification at the LHC trigger: a physics-inspired linear transformer that uses 80% less memory than full attention at matching accuracy, and a hierarchical patch-attention model with geometric message passing that reaches state-of-the-art accuracy among resource-constrained taggers (NeurIPS 2026)
- Built mixture-of-experts particle transformers and showed that routing each particle to a single expert beats a dense model of the same compute on a 188-class jet classification benchmark
- Explained what jet-tagging transformers learn: showed that their attention is nearly binary (each particle attends to one other) and traced the cause to the attention mechanism itself; rewrote the hidden layers of a deployed tagger as if-statements on physical jet quantities that reproduce the network's decisions on 94% of jets
- Deployed transformer models on FPGAs for the CMS Level-1 trigger: quantization-aware training and high-level synthesis to bit-exact RTL, reaching sub-100 ns inference within 10% of the device's logic budget
- Built a hardware-aware neural architecture search that uses learned models of FPGA resource use and latency to find designs that fit the trigger budget (AutoML 2026)
- Trained models on 100-million-particle datasets with distributed training at Argonne National Laboratory; searched LHC data for rare dark matter interactions with an automated analysis framework

Amazon Web Services, Fundamental AI Research New York, NY — September 2025 – December 2025
Applied Scientist Intern

- Tested whether automatic prompt optimizers (GEPA, MIPROv2, APE), which rewrite an LLM's prompt from its own feedback, improve reasoning or only change output formatting and inject memorized facts

- Evaluated each optimized prompt over 10–30 repeated runs with statistical tests (paired t-test, Mann–Whitney U, McNemar's) across GPT-OSS, Claude Sonnet and Qwen3 on AIME math, BigBench Extra Hard and HotpotQA
- Found that prompt optimization gave no significant accuracy gain on 16 of 24 BigBench tasks; of the 8 that improved, 5 were explained by output formatting and 2 by memorized facts in the prompt, leaving 1 task with a reasoning gain
- Showed by ablation that the memorized facts were the source of the gains: removing them from an optimized prompt dropped accuracy back to that of a prompt with formatting changes only
- Tested the hypothesis that the optimizers failed on math because the training sets mix problem types: split the data into single-type subsets (algebra, counting, geometry) and found no improvement there either (0 of 272 GEPA-generated prompts), ruling out dataset diversity as the cause

University of Washington Particle Physics Group Seattle, WA June 2020 – July 2022
Research Assistant
- Applied state-of-the-art machine learning models to particle jet classification, reaching 99% accuracy
- Ranked input-feature importance for jet classification by pruning features on benchmark datasets
- Built autoencoder and GAN models to detect anomalous events at the LHC trigger
- Quantized recursive and recurrent neural networks for FPGA deployment with hls4ml

Pacific Northwest National Laboratory Richland, WA July 2021 – September 2021
Machine Learning Research Intern
- Built few-shot models in PyTorch to predict the solubility of materials
- Analyzed model errors by molecule type and related them to the underlying chemistry
- Cut the input features by 30% with no loss in accuracy, using Spearman correlation to rank feature importance

University of Washington Quantum Materials Lab Seattle, WA March 2019 – July 2022
Research Assistant
- Grew quantum material crystals by flux, vapor and Bridgman methods, including multiple dopings of MnBiTe and $PdTe_2$
- Measured resistivity versus temperature on prepared samples
- Designed and refined growth recipes for new materials with graduate students

IRIS-HEP New York, NY January 2021 – June 2022
Research Fellow
- Extended the Reproducible Open Benchmarks (ROB) platform, a system for comparing physics analyses on shared data, to run Jupyter notebooks; built with NYU computer science and physics faculty
- Built demos introducing ROB to the particle physics community

Washington Nanofabrication Facility Seattle, WA June 2019 – September 2019
Summer Research Fellow
- Fabricated cantilevers in a clean room, designing etch depths and masking layers and verifying them experimentally
- Used the cantilevers to measure fundamental frequencies and study structural transitions in quantum materials

PROJECTS

Distilling Jet Taggers into Physics Equations github.com/aaronw5/jet-tagger-distillation
- Rewrote the latent representations of two jet taggers, the FPGA-deployable real-time model JEDI-linear and the full-size Particle Transformer (ParT), as if-statements on jet physics quantities such as jet and subjet masses
- For JEDI-linear, the distilled formulas reproduce the network's decision on 94% of jets (64 particles per jet) with equal classification accuracy
- The same formulas explain 92% of each neuron's variance in the trained network versus 16% in an untrained copy

Interpretability Explorer for a Trigger-Level Jet Tagger github.com/aaronw5/jedi-linear-explorer
- Wrote every neuron of JEDI-linear, a quantized real-time jet tagger, as an exact formula of its particle inputs (p_T, $\Delta\eta$, $\Delta\phi$) and matched each to a jet physics quantity
- Built an interactive explorer with sparse autoencoders, transcoders and decision trees distilled from the network, evaluated on 50,000 held-out jets

Machine Learning for Cryogenic Transistor Modeling github.com/aaronw5/Cryo-Transistor-ML
- Trained neural-network surrogates of transistor current–voltage curves to extract compact-model (BSIM4) parameters for an open-source 130 nm CMOS process at cryogenic temperature (77 K), validating every fit in circuit simulation
- Lowered fitting error relative to the published parameters on all 18 test devices (combined relative RMS error 0.260 to 0.214)
- First-author paper at the Sim2Sci Workshop @ NeurIPS 2026

SELECTED PUBLICATIONS

Full list, including CMS Collaboration papers: Google Scholar

1. **Wang, A.**, Zhao, Z., Xia, A., Sun, C., Gandrakota, A., Ngadiuba, J., Cavanaugh, R., & Duarte, J. (2026). Patch Hierarchical Attention Transformer for Efficient Particle Jet Tagging. ***NeurIPS 2026***. arXiv:2605.21789.

2. **Wang, A.**, Madan, N., Sobal, V., Trager, M., Kleinman, M., Mansimov, E., Xia, W., & Soatto, S. (2026). On the Clock: Towards Punctual and Productive Time-Budgeted AI Agents. ***Resource-Aware Agentic AI Workshop @ NeurIPS 2026***. arXiv:2610.10833.
3. **Wang, A.**, et al. (2026). Surrogate Assisted Compact Model Extraction for Commercial CMOS Process Operating at Cryogenic Temperatures. ***Sim2Sci Workshop @ NeurIPS 2026***.
4. Weitz, J., Demler, D., Hawks, B., **Wang, A.**, Tran, N., & Duarte, J. (2026). SNAC-Pack 2.0: Scaled-Out Surrogate Neural Architecture Codesign. ***AutoML 2026*** (Oral). arXiv:2605.16138.
5. Pendiyala, K., Zia, H., Lee, T., Legge, T., De Leon, A. J., Zhao, Z., **Wang, A.**, Gandrakota, A., Ngadiuba, J., Cavanaugh, R., & Duarte, J. (2026). Conditional Capacity and Routing in Mixture-of-Experts Particle Transformers. *arXiv preprint* arXiv:2610.02701.
6. **Wang, A.**, Legge, T., Ortiz, J., Limouzi, V., Zhao, Z., Gandrakota, A., Khoda, E. E., Ngadiuba, J., Duarte, J., & Cavanaugh, R. (2025). Why Is Attention Sparse in Particle Transformer? ***Machine Learning and the Physical Sciences (ML4PS) Workshop @ NeurIPS 2025***. arXiv:2512.00210.
7. **Wang, A.**, Zhao, Z., Katel, S., Sahu, V. G., Khoda, E. E., Gandrakota, A., Ngadiuba, J., Cavanaugh, R., & Duarte, J. (2025). Spatially Aware Linear Transformer (SAL-T) for Particle Jet Tagging. *arXiv preprint* arXiv:2510.23641.
8. **Wang, A.**, Gandrakota, A., Ngadiuba, J., Sahu, V., Bhatnagar, P., Khoda, E. E., & Duarte, J. (2024). Interpreting Transformers for Jet Tagging. ***Machine Learning and the Physical Sciences (ML4PS) Workshop @ NeurIPS 2024***. arXiv:2412.03673.
9. Jiang, Q., Palmstrom, J. C., Singleton, J., et al. (2024). Revealing Fermi surface evolution and Berry curvature in an ideal type-II Weyl semimetal. ***Nature Communications*** 15, 2310. doi:10.1038/s41467-024-46633-w.
10. Khoda, E. E., Rankin, D., Teixeira de Lima, R., Harris, P., Hauck, S., Hsu, S.-C., Kagan, M., Loncar, V., Paikara, C., Rao, R., Summers, S., Vernieri, C., & **Wang, A.** (2023). Ultra-low latency recurrent neural network inference on FPGAs for physics applications with hls4ml. ***Machine Learning: Science and Technology*** 4, 025004. arXiv:2207.00559.

Talks and Teaching

Lecture and tutorial, "Transformers," Parts I and II, ML4FP School 2026, Georgia Tech, Atlanta, GA — June 2026

Talk, "Efficient and Interpretable Transformers for Jet Tagging," Fermilab AI Seminar Series, Batavia, IL — March 2026

Talk, "Spatially Aware Linear Transformer (SAL-T) for Efficient Particle Jet Identification," ML4Jets 2025, Caltech, Pasadena, CA — August 2025

Talk, "Fast RNN Inference on an FPGA," APS Division of Particles and Fields Meeting (DPF21), virtual — July 2021

Talk, "Implementation of Jupyter Notebooks into the Reproducible Open Benchmarks for Data Analysis Platform (ROB)," APS DPF21, virtual — July 2021

Talk, "Implementation of Jupyter Notebooks into the Reproducible Open Benchmarks for Data Analysis Platform (ROB)," ML4Jets 2021, virtual — July 2021

Talk, "Jupyter Notebook Compatibility with ROB," IRIS-HEP Fellow Presentations, virtual — June 2021

Posters

"Patch Hierarchical Attention Transformer for Efficient Particle Jet Tagging," NeurIPS 2026 — December 2026

"On the Clock: Towards Punctual and Productive Time-Budgeted AI Agents," Resource-Aware Agentic AI Workshop @ NeurIPS 2026 — December 2026

"Surrogate Assisted Compact Model Extraction for Commercial CMOS Process Operating at Cryogenic Temperatures," Sim2Sci Workshop @ NeurIPS 2026 — December 2026

"Why Is Attention Sparse in Particle Transformer?," ML4PS Workshop @ NeurIPS 2025 — December 2025

"Efficient Transformer Architectures for Jet Tagging," ACAT 2025, Hamburg, Germany — September 2025

"Efficient and Interpretable Transformers for Particle Physics," FastML 2025, ETH Zurich, Switzerland — September 2025

"Interpreting Transformers for Jet Tagging," ML4PS Workshop @ NeurIPS 2024 — December 2024

"Interpreting and Accelerating Transformers for Jet Tagging" (with lightning talk), FastML 2024, Purdue University, West Lafayette, IN — October 2024

Technical Skills

LLM & ML Methods:	GRPO and RL post-training, RLHF, supervised fine-tuning, reward design, trajectory-level data augmentation, RoPE and architecture modification, hardware-aware neural architecture search, LLM agents and evaluation
ML Frameworks:	PyTorch, TensorFlow, Hugging Face, vLLM, SGLang, Megatron, verl, scikit-learn, NumPy, Pandas
Infrastructure:	AWS, multi-node GPU clusters, Docker, Kubernetes, distributed training, ROOT
Hardware ML:	Quantization-aware training (HGQ2, QKeras), hls4ml, da4ml, FPGA resource and latency estimation
Programming:	Python, Bash, Julia, MATLAB, Mathematica, R, Java, LaTeX
Mathematics:	Probability theory, statistics, linear algebra, differential equations, multivariable calculus

MENTORSHIP

Timothy Legge	Transformer interpretability; Ph.D. Physics, UC San Diego	Jan 2025 – Sep 2026
Victor Limouzi	Attention-free transformers; M.S. CS, Imperial College London	Feb 2025 – May 2026
Jacob Ortiz	Attention-free transformers; AI Builders Fellow, M.S. Data Science, UC Berkeley	Feb 2025 – May 2026
Trevin Lee	Mixture-of-experts particle transformers	Aug 2025 – Present
Haris Zia	Switch particle transformers	Aug 2025 – Present